

CLOUD INFRASTRUCTURE SECURITY & HARDENING

FinTrust | Executive-Technical Case Study

ROLE Cloud Engineer - Team D	ENVIRONMENT Ubuntu cloud host • Dockerized financial application • PostgreSQL • Wazuh • Tailscale
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THE CHALLENGE

FinTrust was a cloud-hosted financial application environment that required a stronger defensive foundation before production use. My responsibility was to secure and maintain the cloud host, reduce infrastructure exposure, control administrative access, integrate centralized monitoring, validate implemented controls, and document the environment for the wider security team.

MY RESPONSIBILITY

Assess

Established a host baseline, reviewed listening services, access paths, permissions, and security posture.

Harden

Strengthened SSH and host configuration while preserving application availability.

Control

Implemented role-based access and least privilege for Cloud, SOC, GRC, and Audit functions.

Monitor

Integrated the host with Wazuh and enabled security visibility across configuration, files, and logs.

Validate

Used technical checks, Lynis, Wazuh evidence, and documented verification steps to confirm controls.

Communicate

Translated residual technical risk into remediation priorities and business impact.

SECURITY ARCHITECTURE

Authorized Team Members

↓ Tailscale private connectivity

Ubuntu Cloud Host

SSH hardening • RBAC • least privilege • audit visibility

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Dockerized FinTrust Application

Application services + PostgreSQL

Centralized Security Monitoring

Wazuh Agent → Wazuh Manager / SOC

FIM • SCA • Docker/Application Logs • Auditd

Assess → Harden → Control Access → Monitor → Validate → Communicate Risk

SECURITY CONTROLS IMPLEMENTED

Building a stronger defensive foundation across the host, access layer, connectivity, and monitoring

01 HOST HARDENING

- Captured an initial security baseline and reviewed exposed services and listening ports.
- Hardened SSH: reduced authentication attempts, disabled X11 forwarding, introduced session controls, and enforced key-based access through the final override configuration.
- Reviewed unnecessary exposure and permissions, including SUID/SGID considerations.
- Used Lynis to validate host posture and identify residual hardening opportunities.

02 IDENTITY & ACCESS CONTROL

- Created functional groups for Cloud Admin, SOC, GRC, and Audit responsibilities.
- Applied least privilege and separation of duties instead of giving broad administrative access.
- SOC users operated without unnecessary sudo privileges.
- GRC users received narrowly scoped verification commands to inspect controls without changing the environment.

03 SECURE CONNECTIVITY

- Implemented Tailscale to provide private administrative connectivity between authorized team members and the cloud environment.
- Reduced dependence on direct public administrative access and supported controlled team collaboration.
- Validated connectivity as part of ongoing host operations.

04 SECURITY MONITORING

- Integrated the Ubuntu host with the Wazuh manager for centralized monitoring.
- Used File Integrity Monitoring (FIM) to provide visibility into changes to monitored files.
- Used Security Configuration Assessment (SCA) to identify endpoint configuration weaknesses.
- Collected security-relevant host, audit, Docker, and application telemetry for SOC visibility.

VALIDATION MATTERED

The project was not treated as a configuration exercise. Controls were verified through command output, Wazuh monitoring evidence, configuration review, before/after checks, and written verification guidance for other security functions. Lynis moved from a hardening index of 58 to 59; rather than presenting that small change as a major outcome, the important result was targeted risk reduction while maintaining service availability and identifying what still required remediation.

SKILLS DEMONSTRATED

Linux Hardening • IAM & RBAC • Least Privilege • Wazuh SIEM • FIM • SCA • Docker Security • Tailscale • Lynis • Auditd • Security Validation • Technical Documentation

VALIDATION, RESIDUAL RISK & BUSINESS IMPACT

Security is not complete when controls are deployed - the remaining risk must be understood and communicated.

WHAT THE INFRASTRUCTURE WORK ACHIEVED

Stronger Host

Reduced avoidable infrastructure exposure and strengthened administrative access.

Controlled Access

Introduced RBAC, least privilege, and separation of duties across team functions.

Better Visibility

Centralized monitoring provided evidence across files, configurations, audit activity, and application-related logs.

Repeatable Verification

Documentation and verification commands allowed SOC/GRC teammates to confirm controls independently.

RESIDUAL RISK

The infrastructure controls established a stronger defensive foundation, but the broader Team D assessment identified critical application-layer weaknesses. Those findings were outside the scope of the cloud host hardening alone and meant the environment could not yet be considered production-ready.

Infrastructure controls I owned	Broader application risks identified by Team D
Host hardening RBAC / least privilege Secure connectivity Centralized monitoring FIM / SCA / audit visibility Validation & documentation	Authorization weaknesses Input/query validation gaps Token/session handling concerns TLS gaps Application-layer detection gaps

BUSINESS IMPACT

If the unresolved application weaknesses were exploited in production, the organization could face unauthorized financial transactions, exposure of sensitive customer information, financial loss, compliance consequences, and damage to customer trust and reputation.

RISK-BASED REMEDIATION ROADMAP

IMMEDIATE	Critical authorization, input/query validation, and TLS-related issues requiring urgent attention.
WITHIN 30 DAYS	Strengthen account/session controls, transaction validation, and application-focused monitoring/detection.
WITHIN 90 DAYS	Expand audit coverage, incident-response playbooks, patch cadence, and repeatable authorization testing.

FINAL SECURITY POSITION

FinTrust was not ready for production in its current state. The defensive foundation had been strengthened, but critical application-layer risks still required remediation before the environment could safely handle real customer transactions.

Portfolio case study | Uyo V. Alhassan - Cloud Engineer, Team D. Project details are presented to distinguish individual cloud/infrastructure responsibilities from findings produced by the broader assessment team.